Thickness design in this method utilizes design curves that were developed from numerous runs of the DAMA analysis tool; however, for ease of use, significant simplification has reduced the number of input values required in this method.

All pavement designs for GA airports are assumed to be accomplished for a MAAT = 60°F (15.5°C), so no climate input is required. Analysis is for a single-wheel load and the required input is maximum aircraft weight. The number of departures and other load-related information is standardized for this design procedure. For material properties, only three combinations of pavement materials are considered in this design, further simplifying the design procedure:

- Full-depth asphalt concrete
- 2-inch (50 millimeters) HMA-over-emulsified-asphalt base
- HMA over a 6-inches (150 millimeters) layer of unbound granular base

Design thicknesses contained in the GA design charts are based on main-gear wheel loads equal to 47.5 percent of the gross weight. Above a gross weight of 30,000 pounds (13,500 kilograms), equivalent single-wheel loads were determined for dual-tire gears. A minimum total thickness of 4 inches (100 millimeters) of asphalt structure is recommended for these pavements, regardless of subgrade strength.

Altogether, the design curves developed for GA airports offer an easy-to-use design system based on mechanistic-empirical design principles tailored for use at general-aviation airports.

Air-carrier Airports

Air-carrier airport pavements accommodate aircraft of 60,000 pounds (27,000 kilograms) gross weight or more (Figure 6.4). The comprehensive design manual MS-11, details the air-carrier airport design method, and it is incorporated into SW-1.

The air-carrier design method uses pavement design curves developed from numerous runs of a computer-based M-E analytical model. To account for the differences between aircraft gear and wheel geometry, design aircraft passes are converted to strain repetitions of an equivalent standard aircraft.

The design curves allow variation in MAAT from 40°F (4.4°C) to 75°F (24°C).

The air-carrier design method considers only full-depth asphalt pavements, and no consideration is given to pavement sections utilizing granular base materials. This conforms to Federal Aviation Administration (FAA) design requirements for stabilized base materials for design aircraft over 100,000 pounds (45,000 kilograms) gross weight.

The basic steps of the air-carrier pavement design procedure are as follows:

1. Climate—establishing the appropriate MAAT for the project
2. Traffic—identifying design aircraft and number of departures during the design period
3. Subgrade—describing subgrade strength in terms of resilient modulus
4. Solution—working through the curve (or software) to establish the design thickness for a full-depth asphalt section.



Figure 6.4 Commercial Passenger Aircraft

The Asphalt Institute design method using MS-11 or SW-1 is an authorized alternative to using the FAA pavement design method, subject to approval by FAA regional officials.

6.1.5 Heavy Wheel Loads

Design of HMA pavements for port and industrial facilities with heavy wheel loads is based on the methods detailed in Asphalt Institute Manual MS-23.

Heavy wheel-load vehicles include log-hauling trucks, dump-body haulers, forklift trucks, straddle carriers, rubber-tire hoists, and many special vehicles having four or more tires (Figure 6.5). Heavy-load vehicles can reach gross vehicle weights in excess of 400,000 pounds (180,000 kilograms). Unlike the standardization found in the highway trucking industry, there is little standardization among heavy-vehicle manufacturers for port and industrial applications. Consequently, wheel geometry can vary greatly from one vehicle to another.

The heavy wheel-load design curves published in MS-23 were developed by presolving numerous problems using DAMA. Due to the expected low number of design vehicle passes, HMA pavements in port and industrial applications are not expected to fail from fatigue cracking. The design curves are based only on subgrade rutting.

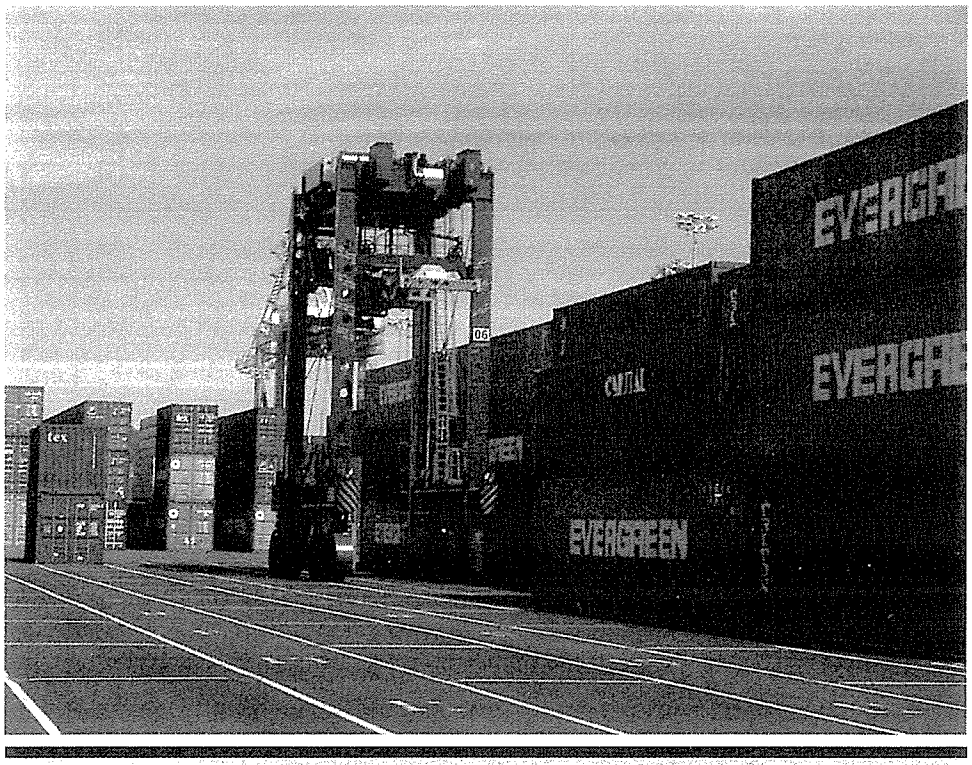


Figure 6.5 Straddle Carrier at Port Facility

A further simplifying assumption is that only full-depth asphalt materials are considered for heavy wheel applications.

Climate variability is greatly simplified for heavy wheel-load pavements in that only two climate options are available:

- MAAT less than or equal to 55°F (13°C)
- MAAT greater than 55°F (13°C)

Essentially, the choice is between a cold climate, where the subgrade is subjected to freeze hardening and thaw weakening, and a warm climate with no freeze/thaw concerns.

An important aspect of understanding the design vehicle loads and wheel geometry is to ascertain whether the problem can be considered a single-wheel, dual-wheel, or multiple-wheel design problem. For single-wheel loads, subgrade rutting is analyzed only at the point directly below the wheel. On the other hand, dual- and multiple-wheel loads can superimpose at depth, complicating the process of determining the location and magnitude of the maximum subgrade compressive strain. If a distance greater than eight times the radius of a loaded contact area separates the two wheels, the problem may be considered a single-wheel problem.

Single-wheel load design problems are relatively straightforward, requiring only the magnitude of the wheel load, the tire-ground contact pressures, the contact area, and the number of load repetitions. Dual- and multiple-wheel problems are more complicated, requiring additional

wheel geometry data information. Dual- and multiple-wheel problems can then be reduced to an equivalent single-wheel load for the specified load geometry.

The design process for dual- and multiple-wheel vehicles is described in detail in MS-23. In addition, SW-1, greatly simplifies the heavy wheel load design process for the user.

6.1.6 Planned Stage Construction

Planned stage construction is the construction of roads and streets by applying successive layers of asphalt concrete according to design and a predetermined time schedule. Two possible circumstances where stage construction is advantageous are the following:

- When adequate funds are not immediately available for a full design, but anticipated at some later date
- When long-term traffic projections (20–25 years) are not considered reliable. In this situation, traffic estimates for the second-stage construction can be made while the facility is in use after first-stage construction.

The procedure is based on the assumption that the second stage will be constructed before the first stage shows serious signs of distress. Generally, it is held that the first stage should be designed such that 60 percent of the pavement's remaining life is used during the first stage, leaving 40 percent to be used up during the second stage.

Planned stage construction should not be confused with major maintenance or rehabilitation of existing pavements. Readers are referred to MS-1 for further details about this method and example problems.

6.1.7 Overlay Design

Design of HMA overlays is based on either the effective thickness or the deflection method, as detailed in Asphalt Institute Manual MS-17, *Asphalt Overlays for Highway and Street Pavements.*

Effective Thickness Method

The effective thickness procedure for structural evaluation and overlay design is based on the concept that pavements deteriorate over time as a result of load repetitions imposed by traffic. Furthermore, as a pavement uses part of its total life, it behaves as if it were an increasingly thinner pavement. Consequently, the "effective" thickness of an aging pavement is always less than its actual thickness.

The effective thickness of an existing pavement layer can be estimated from its pavement condition and knowledge of its composition and actual thickness. Conversion factors ranging from 0.0 to 1.0 are assigned to the layer based on its composition and condition. The actual thickness of the pavement

layer multiplied by the conversion factor yields the effective thickness. Conversion factors are found in Table 8–1 of MS-17.

For example, if the conversion factor for rubblized concrete has been established as 0.5, then a 10-inch rubblized concrete layer would have an effective thickness of 5 inches of HMA for calculating overlay thickness.

In order to calculate the overlay thickness, the designer must first calculate the design thickness for a new full-depth asphalt pavement using applicable input values for climate, traffic, and subgrade strength. Subtracting the effective thickness of the existing pavement from the design thickness for the new pavement yields the required thickness of overlay. A simple equation to describe the overlay thickness calculation is as follows:

$$T_o = T_n - T_e$$

Where

T_o = thickness of overlay
T_n = thickness of a new pavement for expected traffic, climate, subgrade inputs
T_e = effective thickness of existing pavement

Maximum Deflection Method

The Asphalt Institute deflection procedure for calculating overlay thickness is based on the representative rebound deflection as measured by the Benkelman Beam or correlated from other vibratory or impulse-based nondestructive testing devices. A full description of the method is beyond the scope of this handbook, and the reader is referred to MS-17 for a detailed description of the procedure. These two methods of overlay design have been incorporated in the SW-1 software.

6.2 Relationship to Other Design Methods

The Asphalt Institute recognizes that other flexible-pavement design methods are available to engineers. These other design procedures fall into two basic categories: empirical and mechanistic-empirical. A separate discussion on Perpetual Pavement design is included in this section.

6.2.1 Empirical Methods

Empirical methods are based entirely on the experience of evaluating well-performing and poorly-performing pavements along with specifically constructed test sites such as the AASHO test road. The commonly used CBR method is an empirical method developed in the 1920s by members of the California Department of Highways from experience with its pavements. It was validated with field testing and improved for use on airfields during World War II and is still in wide use today.

The 1993 AASHTO Guide is an empirical method derived from the AASHO Road Test of the 1950s. While the method is widely used and highly respected, the AASHO Road Test pavements were constructed of relatively uniform mixes, built on a single subgrade soil, and exposed to a single environment—Illinois. These and other facts anchor the validity of the data to the specific mixes, soil types, and climatic conditions experienced in Illinois in the 1950s and open the door to criticism of its validity in other conditions.

Empirical methods are limited by the depth and breadth of the data set. For example, it is well established that HMA stiffness varies with pavement temperature, yet the CBR method makes no allowance for HMA stiffness variation due to monthly or seasonal climate variation. Likewise, the AASHTO method has similar difficulty in transferring the experience of one climate to that of another. For these reasons, empirical methods, while convenient, may not make the best predictions for pavement thickness in other locations or for other conditions.

6.2.2 Mechanistic-Empirical Methods

The Asphalt Institute methods are based on mechanistic-empirical principles and are derived from work done in the 1970s and 1980s by the Asphalt Institute staff, member company engineers, and a team of consultants consisting of Mr. Fred N. Finn, Dr. Carl L. Monismith, and Dr. Matthew W. Witczak. These experts established fatigue and rutting criteria that are still in use today in the AI methods and others. Dr. Witczak developed the

computer-based M-E analytical model, DAMA, which was used to create the Asphalt Institute design curves published in MS-1 and other manuals. These efforts established the Asphalt Institute as a pioneer in M-E pavement design methods at a time when the concept was relatively new.

In the past few decades M-E design methods have become widely accepted. It can be said that M-E-based flexible-pavement design is on the path to becoming the standard rather than the exception. Several state highway departments have developed their own mechanistic-empirical design systems including *WINFLEX*, *MnROAD*, and *EverPAVE*, the official design methods of the Idaho, Minnesota, and Washington State Departments of Transportation, respectively. Air-carrier airport-pavement designers are able to tap the power of M-E methods in *LEDFAA*, a computer program developed by the FAA. *PCASE* is the M-E design method applicable to military airfields operated and maintained by the U.S. Department of Defense.

At the time of this writing, a new national M-E Pavement Design Guide (MEPDG) has been developed under sponsorship of the National Cooperative Highway Research Program and is in the process of being introduced to the state highway departments by the Federal Highway Administration. It is widely expected that the MEPDG will be adopted by AASHTO as its pavement design guide at some time in the future.

As discussed in the next section, the Perpetual Pavement design concept, based entirely on M-E principles, is a pavement design strategy that is gaining support across the country.

The Asphalt Institute M-E design method is now part of an increasingly populated field of powerful pavement design tools available to civil engineers worldwide. The reader is referred to Tables 6.1 and 6.2 for a comparison of some of the features of a select group of M-E methods for highway and airport applications.

6.2.3 The Perpetual Pavement Method

The basic concept of a Perpetual Pavement is that through careful mix type and material selection together with high-quality construction, a pavement structure can be built that is resistant to permanent deformation and fatigue cracking. Surface-initiated distress, such as top-down cracking and weathering/raveling, may still occur, but through a regular process of milling and overlaying, the pavement structure never deteriorates. Like the roof on the typical residential structure, the surface layer will need to be replaced periodically, but the underlying pavement structure remains solid and strong.

The Perpetual Pavement design concept consists of the following layers:

1. An unyielding subgrade forms the pavement foundation.
2. The base layer consists of a flexible and fatigue-resistant HMA mix—possibly a "rich" mix with a slightly higher binder content.
3. Intermediate layers consist of stiff, rut-resistant mixes such as Superpave or stone matrix asphalt (SMA).

Table 6.1

Comparison of Selected Mechanistic-Empirical Highway Pavement Design Methods

	Agency	Asphalt Institute		National Cooperative Highway Research Program	Asphalt Pavement Alliance	Minnesota Department of Transportation	Idaho Transportation Department	Washington State Department of Transportation
	Method	DAMA	*MS-1*	*MEPDG*	*PerRoad*	*MnPAVE*	*WINFLEX*	*EverPAVE*
	Mechanistic Model	Chevron N-Layer	Chevron N-Layer[1]	JULEA / DSC2D	WESLEA	WESLEA	Chevron N-Layer	WESLEA
Inputs	Traffic	Wheel Load	ESAL	Load Spectra	Load Spectra	ESAL or Load Spectra	ESAL	ESAL
Inputs	Subgrade	M_r	M_r	M_r	M_r	M_r	M_r	M_r
Inputs	Materials	E*	E*	E*	E*	E*	E*	E*
Inputs	Climate	Mean Monthly Air Temp	Mean Annual Air Temp	Enhanced Integrated Climate Model	Mean Seasonal Air Temperature (5 seasons)	Mean Seasonal Air Temperature (5 seasons)	Mean Seasonal Air Temperature (4 seasons)	Mean Seasonal Air Temperature (4 seasons)
Inputs	Other	Allows up to Five Layers	Six Pavement Models	Allows to up to Ten Layers	Allows up to Five Layers	Allows up to Five Layers	Allows up to Four Layers + Overlay	Allows up to Four Layers + Overlay
Failure Criteria	HMA Fatigue	AI Equation	AI Equation[1]	AI Equation	Various	AI Equation	AI Equation[3]	WSDOT Equation
Failure Criteria	Subgrade Rutting	AI Equation	AI Equation[1]	MEPDG Equation	Various	Illinois Equation	AI Equation[3]	AI Equation
	Output	Remaining Life	HMA Thickness	Remaining Life	Remaining Life	Remaining Life / HMA Thickness	Overlay Thickness	Damage Levels / Overlay Thickness
	Remarks		[1] Pertains to DAMA	Other Criteria: Top-down Cracking Thermal Cracking IRI	User can specify criteria for the top, middle, and bottom of each layer	*MnPAVE* has a layer thickness goal-seeking option	[3] *WINFLEX* allows user to select other equations	

Table 6.2
Comparison of Selected Mechanistic-Empirical Airport Pavement Design Methods

Agency		Asphalt Institute		Federal Aviation Administration	US Army Corps of Engineers
Method		DAMA	MS-11 *Airport Design Curves*	*LEDFAA (FAArfield)*	*PCASE*
Mechanistic Model		Chevron N-Layer	Chevron N-Layer[1]	LEAF	JULEA
Inputs	Traffic	Wheel Load	Number of Aircraft Departures	Number of Aircraft Departures	Design Aircraft Passes
	Subgrade	M_r	M_r	M_r	M_r
	Materials	E*	E*	E*	E*
	Climate	Mean Monthly Air Temperature	Mean Annual Air Temperature	N/A	Mean Monthly Air Temperature
	Other	Pavement Structure (up to Five Layers)	Full Depth Asphalt	Pavement structure	Pavement structure (up to Five layers)
Failure Criteria	HMA Fatigue	Asphalt Institute Equation	Asphalt Institute Equation[1]	Heukelom and Klomp Equation	Heukelom and Klomp Equation
	Subgrade Rutting	Asphalt Institute Equation	Asphalt Institute Equation[1]	FAA Equation	USACE Equation
Output		Remaining Life	HMA Thickness	Remaining Life / Pavement Structure Thickness	HMA Thickness
Remarks			[1] Pertains to DAMA	E* fixed at 200,000 psi	

4. The wearing course is a high-quality Superpave, open-graded friction course, or SMA intended to be milled and replaced at regular intervals.

The thrust of Perpetual Pavement design is to combine a rut-resistant, impermeable, and wear-resistant top structural layer with a rut-resistant and durable intermediate layer and a fatigue-resistant and durable base layer. Though a Perpetual Pavement is focused at high-volume traffic roadways, the justification may be made for medium- and low-volume roads as well.

Essential to the Perpetual Pavement concept is the assumption that the subgrade soils form a strong foundation. Improvements to drainage, soil stabilization, or over excavation and replacement may be required to create the proper foundation for a successful Perpetual Pavement.

The HMA layers in a Perpetual Pavement may include a "rich" bottom layer with a slightly higher-than-typical binder content to create additional fatigue resistance. The intermediate layers are intended to be stiff rut-resistant mixes, such as the Superpave mixes that are currently used in the United States. The mixes used for the wearing course can vary depending on the specific needs of the area. Superpave mixes provide additional rut-resistance and a smooth riding surface. SMA mixes might be selected for exceptional durability on high-volume or heavily loaded streets and highways. An open-graded friction course (OGFC) may be chosen to reduce splash and spray and/or reduce pavement noise while providing a smooth-wearing surface. In a Perpetual Pavement, the wearing course is intended to be milled and overlaid at regular intervals to remove surface distresses and maintain smoothness.

At the core of the Perpetual Pavement design method is the concept of an "endurance limit." This concept holds that there is a level of strain below which fatigue damage does not occur for any number of load repetitions. For the condition of fatigue at the bottom of the lowest HMA layer, an endurance limit of 70 microstrain has been established. Long-term research studies will validate this limit. In the life of a Perpetual Pavement, horizontal tensile strain at the bottom of the lowest asphalt-bound layer does not contribute to fatigue failure if it does not exceed the 70 microstrain threshold.

Similarly, a threshold of 200 microstrain is currently understood to be the limit for vertical compressive strain in subgrade materials. It is held that a strain level below 200 microstrain does not contribute to subgrade rutting in Perpetual Pavement design.

Perpetual Pavement design is conducted using M-E design principles for layered elastic materials using the *PerRoad Perpetual Pavement Software.* The cumulative-damage factor concept is applied to both fatigue and rutting conditions, but rather than allow pavement damage to reach 1, damage in a Perpetual Pavement design is never allowed to exceed 0.1.

The reader is referred to Table 6.1 for a comparison of *PerRoad* to the Asphalt Institute's and other methods and to Table 6.3 for information on how to acquire the software and other information regarding Perpetual Pavement design.

6.3 Design Manuals and Software

As discussed throughout this chapter, the Asphalt Institute has developed design manuals and software that have gained wide acceptance among pavement designers in the United States and throughout the world. Table 6.3 lists pavement-design manuals available from the Asphalt Institute. These publications and software may be obtained by contacting the Asphalt Institute at the following address:

Asphalt Institute
2696 Research Park Drive
Lexington, Kentucky 40511-8480

www.asphaltinstitute.org

Table 6.3
Asphalt Institute Design Manuals and Software

Application	Manual No.	Manual Title
General Aviation Airports	IS-154	*Thickness Design—Asphalt Pavements for General Aviation*
Highways, Roads, and Streets	IS-181	*Asphalt Pavement Thickness Design—A Simplified and Abridged Version of MS-1*
Parking Lots, Service Stations, and Driveways	IS-91	*Full Depth Asphalt Pavements for Parking Lots, Service Stations, and Driveways*
Highways, Roads, and Streets	MS-1	*Thickness Design—Asphalt Pavements for Highways & Streets*
Air-Carrier Airports	MS-11	*Thickness Design—Asphalt Pavements for Air Carrier Airports*
Overlay Design	MS-17	*Asphalt Overlays for Highway and Street Rehabilitation*
Ports and Industrial Yards	MS-23	*Thickness Design—Asphalt Pavements for Heavy Wheel Loads*
All	SW-1	*Asphalt Pavement Thickness Design for Highways, Airports, Heavy Wheel Loads, and Other Applications*

The Asphalt Institute recognizes that it is not alone in publishing thickness design procedures for asphalt pavements. In Table 6.4, we list a selected group of excellent design methods for highways, airports, military pavements, and Perpetual Pavement design. While we have attempted to list those methods that may have the widest possible application, we also recognize that space limitations make it impossible to list all flexible-pavement design methods available.

Table 6.4
Contact Information for Selected Pavement Design Methods

Agency	Application	Design Method	Mailing Address	Phone	Web site
Asphalt Pavement Alliance	Perpetual Pavements	PerRoad	Asphalt Pavement Alliance 5100 Forbes Boulevard Lanham, MD 20706-44017	877-APA-0077	www.asphaltalliance.com
American Association of State Highway and Transportation Officials	Highways	AASHTO 1993	AASHTO 444 N. Capitol Street NW, Suite 249 Washington, DC 20001	(202) 624–5800	www.aashto.org
Federal Aviation Administration	Airports	LEDFAA	William J. Hughes Technical Center Atlantic City Intl Airport, NJ 08405	(609) 485–5147	www.airporttech.tc.faa.gov
Idaho Transportation Department and University of Idaho	Highways	WINFLEX	Idaho Transportation Department P.O. Box 7129 Boise, ID 83707–1129	(208) 334–8440	www.webs1.uidaho.edu/bayomy/Winflex.htm
Minnesota DOT	Highways	MnPAVE	Office of Materials, MS 645 1400 Gervais Avenue Maplewood, MN 55109–2044	(651) 366–5496	www.mrr.dot.state.us
National Cooperative Highway Research Program	Highways	MEPDG	500 Fifth St. NW Washington, DC 20001	(202) 334–3232	www.trb.org/mepdg
US Army Corps of Engineers	Military	PCASE	Transportation Systems Center 12565 West Center Road Omaha, NE 68144–3869	(402) 221–7260	www.pcase.com
Washington State DOT and University of Washington	Highways	EverPAVE	E & E Materials Laboratory P.O. Box 47365 Tumwater, WA 98504–7365	(360) 709–5470	www.wsdot.wa.gov

6.3.1 SW-1 Asphalt Pavement Thickness Design Software

SW-1 is the Asphalt Institute's Windows-based asphalt pavement thickness design software that provides users a single computer-based design solution for highways, airports, heavy wheel applications, and other applications such as parking lots. It includes the Asphalt Institute pavement design curves for highways, airports, heavy wheel loads, and other applications, which have been digitized for use in SW-1. Also included is the DAMA analysis tool for advanced structural analysis. Tying it all together is a graphical user interface to ease handling of user inputs, pavement design reports, and management of data files.

One of the benefits of SW-1 is a versatile and powerful user interface, as shown in Figure 6.6. From this interface, users collect input data, report output, and manage data files for asphalt pavement design for all types of applications. SW-1 uses a tabular format to clarify the type of input required and to aide the user in stepping through a pavement design.

A design wizard is automatically invoked on the initial setup of a new problem. The wizard steps the user through the process of inputting all the required data—climate, traffic, subgrade, materials. The final step through the wizard leads to a design solution and the report page.

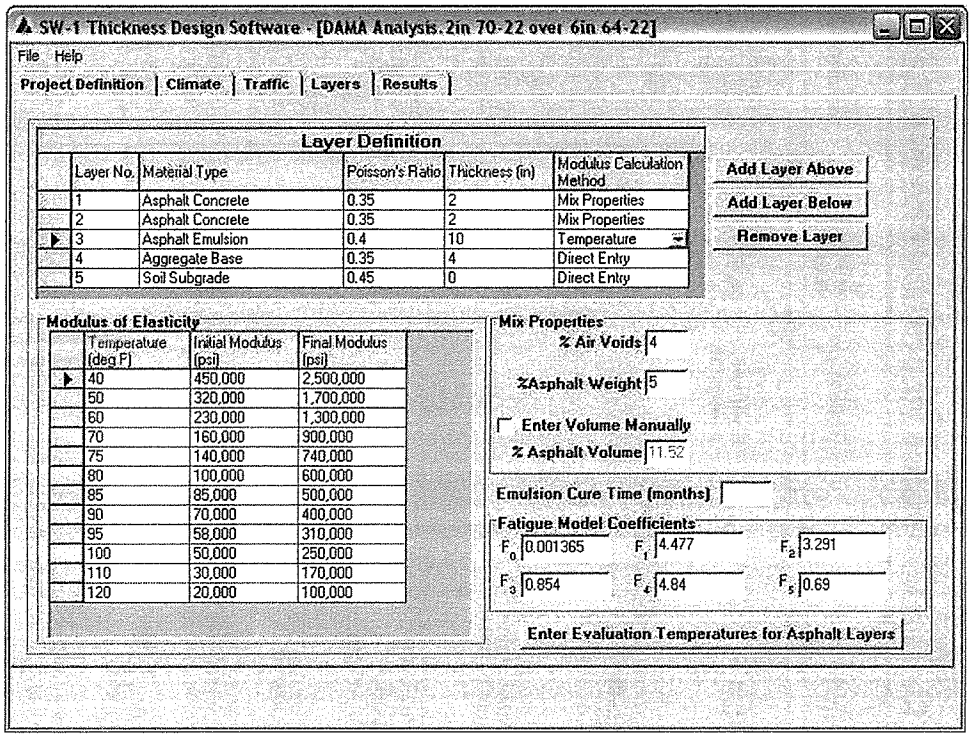


Figure 6.6 Screenshot of SW1, Thickness Design Software

Pavement design solutions are viewed on a professional report page in either summary or detailed versions. The Asphalt Institute logo is prominently displayed on the design report to indicate that the design was accomplished using the Asphalt Institute method.

SW-1 includes an extensive help file that is embedded in the software. In addition, the companion SW-1 *User's Guide* addresses common technical and operational questions. Both the help file and the *User's Guide* include numerous example problems that illustrate how the Asphalt Institute method can be applied to a specific problem.

In summary, SW-1 is a comprehensive design tool that provides users with the power of all the Asphalt Institute design methods (design curves and DAMA) in one convenient and easy-to-use software package. For more information on SW-1, the user is referred to the Asphalt Institute Web site at www.asphaltinstitute.org.

Resources

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CHAPTER 7

MANUFACTURE OF HOT MIX ASPHALT

This chapter provides information on the manufacturing of hot mix asphalt using batch and drum-mix plant facilities. The common features and differences of the two mixing plant types are discussed. Controlling the aggregate gradation and asphalt content is detailed. Mixing and coating with both type plants is covered. Incorporating RAP into hot mix with both plant types is discussed. Finally, a trouble-shooting guide is provided.

Section 7.1 Types of Plant Facilities/Production Processes
- 7.1.1 Batch Facilities
- 7.1.2 Drum-Mix Facilities

Section 7.2 Equipment Common to Both Types of Facilities
- 7.2.1 Liquid Asphalt Storage
- 7.2.2 Aggregate Stockpiling
- 7.2.3 Virgin Aggregate and RAP Cold-Feed Bins
- 7.2.4 Aggregate and RAP Transfer Conveyors
- 7.2.5 Aggregate Dryer
- 7.2.6 Dust Collection Equipment
- 7.2.7 Dust-Return Equipment
- 7.2.8 Mineral Filler Additive Storage and Feed Silos
- 7.2.9 Lime Additive Bulk Storage and Feed Silos
- 7.2.10 Fiber Additive Equipment
- 7.2.11 Finished Product Storage Silos and Load-Out Equipment

Section 7.3 The Batch Plant Production Process
- 7.3.1 Aggregate Gradation Control with a Batch Plant
- 7.3.2 Asphalt Content Control with a Batch Plant
- 7.3.3 Mixing/Coating with a Batch Plant
- 7.3.4 Recycled Mix Production with a Batch Plant

Section 7.4 The Drum-Mix Production Process
7.4.1 The Development of the Drum-Mix Process
7.4.2 Aggregate Gradation Control in a Drum-Mixer
7.4.3 Asphalt Content Control in a Drum-Mixer
7.4.4 Mixing/Coating with a Drum/Continuous Plant
7.4.5 Recycled Mix Production with a Drum-Mixer

Section 7.5 Mix Deficiencies and Problem Solving
7.5.1 Drum-Mix Production Problem-Solving Guide
7.5.2 Batch-Plant Production Problem-Solving Guide

7.1 Types of Plant Facilities/ Production Processes

Two types of plant facilities/production processes for producing hot mix asphalt (HMA) paving mixtures are popular today:

- Batch facilities
- Drum-mix facilities

In each type of facility, aggregates are proportioned according to the requirements of the job-mix formula, dried and heated, and then mixed with the required asphalt binder to create the proper ratio of aggregates and liquid asphalt binder. With both types of facilities, supplemental fine aggregate or liquid additives can be added and reclaimed asphalt pavement (RAP) can be used to make recycled mixes. However, additive procedures are slightly different in each process.

7.1.1 Batch Facilities

The first hot mix asphalt plants were batch facilities. These were popular from the early 1900s until the advent of drum-mixers in the 1970s. Many batch facilities still exist, producing mix daily, although over 80 percent of HMA tonnage in North America is now produced with drum-mix plants. Figure 7.1 shows the key equipment components in a batch facility, and Figure 7.2 shows a cutaway view of the flow process through the batch tower.

In the batch-plant process, individual aggregates are proportioned from the cold feed bins onto a single collector belt and sent to the dryer to be dried and heated. The dried and heated materials flow into a bucket elevator to be carried to the top of the batch tower and deposited onto sizing screens. Multiple screen decks separate the aggregate into different sizes and direct the fractionated aggregate into hot bins. The screens in the tower determine the size of the aggregates in the hot bins and how the blended aggregate mixture from the dryer is fractionated.

Aggregates are then dried, heated, sized, and stored in the hot holding bins, for proportioning according to the mixture specified.

The mixture is made by weighing the fractionated aggregates stored in the hot bins with an accumulative aggregate weigh hopper, positioned below the hot bins. Each predetermined fraction of material is individually weighed into the aggregate weigh hopper until the entire batch is weighed.

In a separate action, asphalt is weighed in a weighing device, typically called an asphalt weigh bucket or asphalt scale.

After the aggregate and the binder are weighed, the aggregates are dropped into the pugmill mixer to be blended together in what is called a

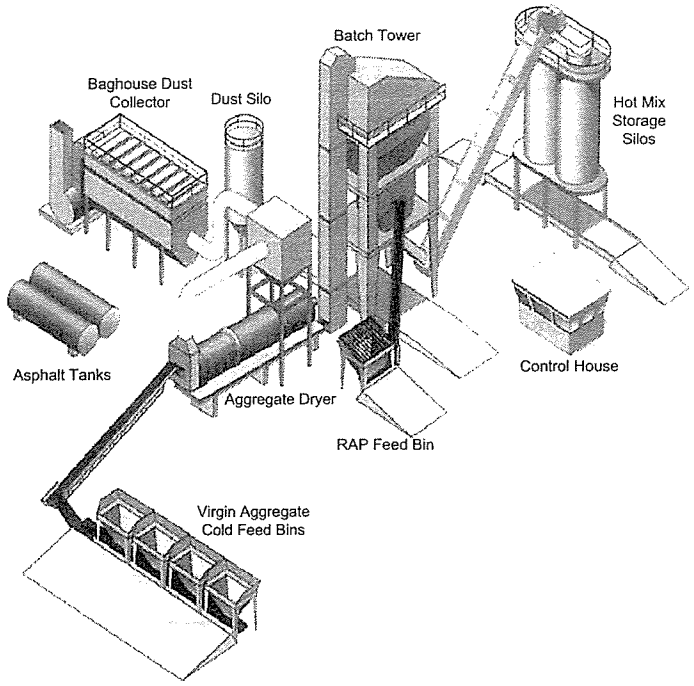


Figure 7.1 Components of a Modern Batch Plant

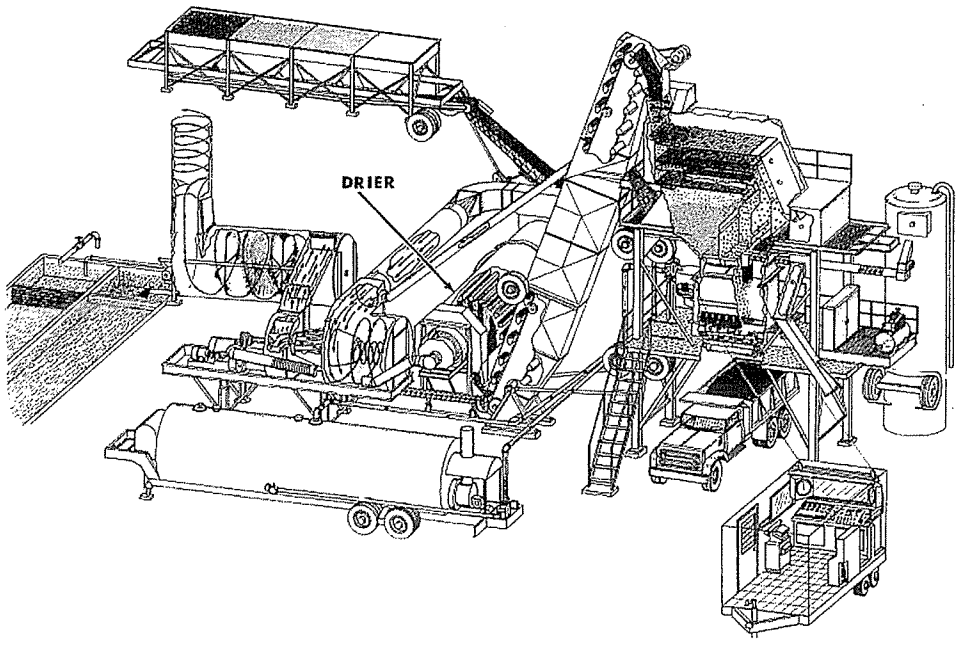


Figure 7.2 Flow of Material through a Batch Plant

dry-mix cycle. This 2- to 5-second cycle blends the aggregates thoroughly before the asphalt binder is added.

When the dry-mix cycle is completed, the asphalt is dropped into the mixer and the wet-mix cycle starts. The wet-mix cycle lasts 25 to 45 seconds, with the usual length being 30 to 35 seconds. The cycle time is determined by the time required to produce a homogenous, well-coated mix. Uniform coating of large-rock mixes typically takes longer than fine mixtures. The time is usually determined subjectively by visual inspection. Some regulatory agencies set minimum times for the mixing process.

7.1.2 Drum-Mix Facilities

Figures 7.3 and 7.4 show the key components and flow process in a drum-mix facility.

The continuous flow of a drum-mix plant requires the availability of stockpiled aggregate material that will meet the specifications when proportioned without rescreening. The individual aggregate materials are proportioned directly from the cold-feed bins to the final job-mix formula. The materials are then sent to the drum to be dried and heated. In route to the drum, the blended aggregate passes over a belt scale, which is a continuous-weighing device. The plant's control system uses the reading from the belt scale to calculate the liquid asphalt binder and other additives

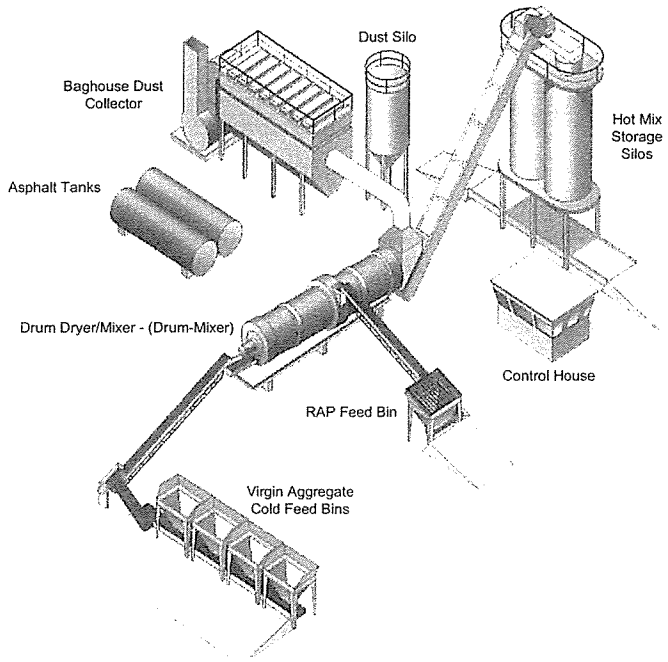


Figure 7.3 Components of a Modern Drum-Mix Plant

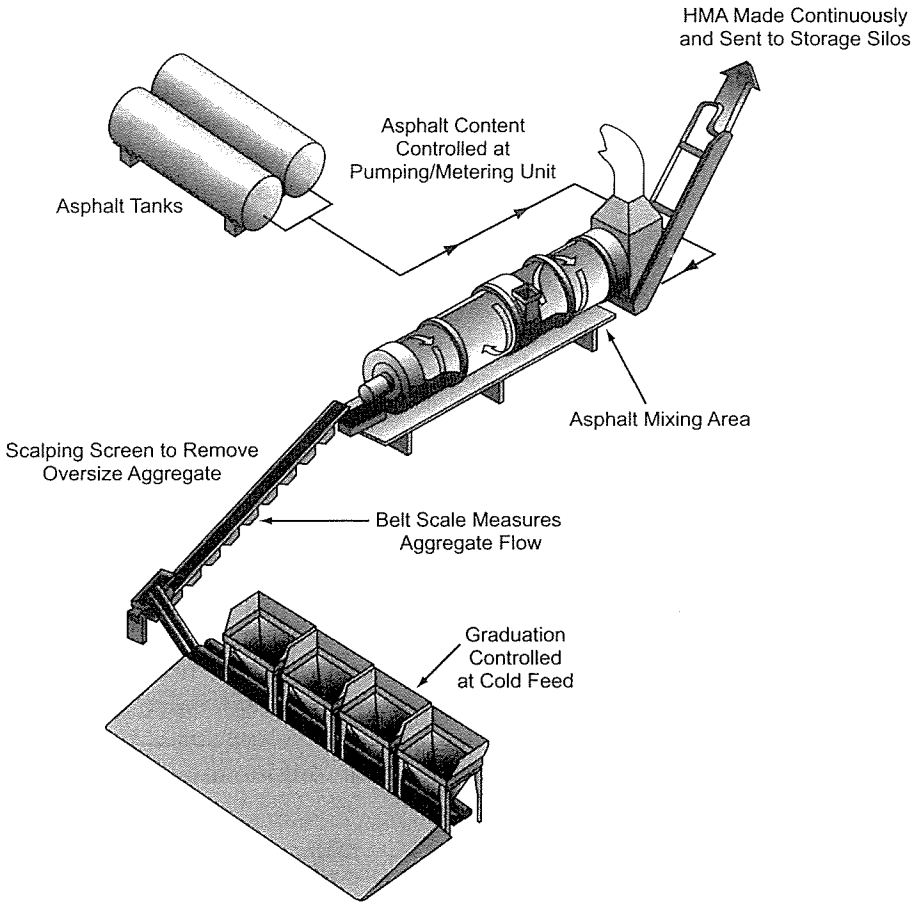


Figure 7.4 Flow Process through a Drum-Mix Plant

required for the mixture. Drum-mix plants can be designed as one-piece drum-mixers or with separate dryer and continuous-mixer units. Both are referred to as drum-mixers. Different styles of drum-mixers are illustrated and discussed later in the chapter.

7.2 Equipment Common to Both Types of Facilities

Regardless of the proportioning and mixing style, both types of facilities have common equipment components that operate similarly and have essentially the same functions:

- Liquid binder storage tanks
- Aggregate and RAP stockpiling
- Virgin aggregate and RAP feed bins
- Aggregate and RAP transfer conveyors
- Aggregate dryers
- Dust collection equipment (from aggregate dryer)
- Collected dust return equipment
- Mineral filler additive storage and positive feed controls
- If required by specifications—lime additive bulk storage and positive feed controls
- Equipment for the addition of fibers
- Finished product storage silos and load-out equipment

Plant automation is similar in both—and typically computer controlled—although blending automation is different because of the differences between the batch and the drum-mix processes. Most automated plant-control systems, whether batch oriented or drum-mixer, provide for multiple stored-mix formulas and automatically handle all the blending, metering, and mixing. All computer-controlled plants can provide cumulative material totals for a daily production report, which aids inventory control.

In both plant types, truck load-out and ticketing systems are also computerized and provide not only weight figures but also detailed project information on the load ticket.

These common equipment groups are discussed below, along with key optimum operating practices for controlling the process and producing consistent high-quality mix with these common components. Specific operational differences between the batch process and the drum-mix process are addressed later in the section.

7.2.1 Liquid Asphalt Storage

Liquid asphalt binder is generally stored inside insulated and heated tanks (see Figure 7.5) adjacent to the other plant equipment. Multiple storage tanks are common at stationary facilities and are not uncommon with portable plants. This allows the producer to keep different grades of asphalt binder properly separated.



Figure 7.5 Vertical Asphalt Storage Tanks

Tanks must be kept heated to maintain the asphalt binder at the proper mixing temperature. Tanks are usually heated with hot oil using a heat transfer oil system that keeps the other plant equipment at operating temperature.

Tanks that store modified binders are often equipped with agitators or stirrers to keep the modified binder constantly circulating. Circulation pumps can be used for the same purpose, even without agitators or stirrers. Some asphalt binders can have specific agitation requirements.

HMA producers are encouraged to consult with the liquid asphalt supplier for proper storage requirements.

Pipes that transfer the liquid asphalt to the mixing area of the plant are insulated and surrounded with an outer pipe or "jacket" containing hot oil from the plant's heating system to keep the binder at the proper temperature throughout the entire process.

Special processing equipment is available to blend mixture additives into the hot asphalt lines or plant mixing area. Antistrip additives, fibers, RAP, or other mixture modifiers are often added after the aggregate has entered the drum. Special processing equipment is also used for mixing crumb-rubber additives and latex into the liquid asphalt for special mixes. This equipment varies by application requirement and is usually computer controlled.

Plant personnel should monitor storage temperatures regularly to make sure the binder is kept at the proper temperature. Thermometers in the tanks and lines are connected to digital displays and graph recorders in the control room for this purpose. Binder temperature affects not only pumping and metering efficiency but also coating characteristics, so ensuring that binder temperature is maintained throughout the process is critical for proper mixing operations. Heater temperatures should be set at required storage temperatures, or not more than 20°–30°F (11°–13°C) above those temperatures. Excessively high temperatures cause the asphalt to build up and harden around the heating elements. These excessively high temperatures can also prematurely age the binder. The recording devices provide visual proof that specification temperature limits have not been exceeded. The objective of storing asphalt binders at an HMA plant is to maintain an adequate supply of material to keep the plant running between binder deliveries. The amount of storage needed at an HMA plant is a function of anticipated production rate, type of binder(s), and the anticipated delivery schedule.

7.2.2 Aggregate Stockpiling

Aggregates are delivered to plant sites by truck, rail, barge, or belt conveyor directly from the aggregate source. Aggregates are usually stockpiled onsite and then fed into cold feed bins with a front-end loader. A plant can also be fed directly from the stockpile by feeders under the stockpiles or by trucks dumping directly into the cold feed bins. The most common approach is to feed the bins from a stockpile with a front-end loader.

Operational concerns for HMA production at the plant site focus on the following key areas.

Cross Contamination

Materials should be kept separated from one another by space or by dividers before being loaded into the cold feed bins for mixing.

Subgrade Contamination

If materials are moved with a front-end loader, the operator must ensure that the bucket does not pick up subgrade materials along with the desired aggregate. Producers usually treat the ground to stabilize it or place a base aggregate-type material below the stockpile to minimize this problem. The latest trend at permanent plant sites is to actually pave the stockpile area with HMA to prevent subgrade contamination and to expedite moisture drainage in the stockpiles.

Wheel and Track Contamination

Driving on the aggregate stockpiles with haul trucks or stockpiling equipment can also cause contamination from materials that stick to the vehicle wheels or tracks, and should be discouraged. The importance of this guideline is relative to local conditions.

Breakdown/Gradation Change

Materials can change in gradation if haul trucks or stockpiling equipment are allowed to drive on and to work the aggregate stockpile. Degradation of aggregate materials is a function of material handling and aggregate characteristics. The gradation of the material can change every time it is handled. A thorough understanding of the aggregate materials and aggregate handling procedures must be accounted for in the design and construction process.

Segregation/Gradation Change

Materials that are not of uniform size are prone to segregation during handling operations. When aggregates are dropped from a significant height, such as from a stacking conveyor, or if aggregates are allowed to flow down the face of a stockpile, segregation can occur. Both practices should be discouraged. Conveyors that rise up and down minimize this problem. Elevated stockpiles do not always have to be constructed using haul trucks and ramps. Available space for stockpiling, however, often dictates compromises in best practices. Materials that are segregated slightly can often be reblended satisfactorily if the loading equipment mixes coarse sides and fine sides from the stockpile face before charging the cold-feed bins. Segregation is probably the single biggest concern in stockpile management at a hot mix plant. If segregation concerns exist, material samples can be taken directly from the discharge of the cold bins to verify whether the gradations of the individual materials have been compromised.

Minimizing Stockpile Moisture

Providing a solid draining surface under the stockpile area not only helps keep the subgrade material from contaminating the cold feed bins but also helps keep aggregates dry. Moisture drastically affects the production of the plant and fuel consumption. If moisture contents in the stockpiles change, the mix temperature will also fluctuate unless the operating characteristics (burner and draft settings) are constantly adjusted. This is very difficult with high-production HMA plants. For these reasons, hot mix